

Riya Singh

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 Portfolio

SUMMARY

Computer Science undergraduate with a strong focus on **AI systems, backend engineering, and scalable system design**, experienced in developing distributed applications integrating Machine Learning models with production-ready APIs. Proficient in **Python, Core Java, MongoDB, and SQL**, with hands-on experience in NLP, RAG systems, multithreading, and socket programming. Skilled in building **scalable backend architectures, concurrent systems, and distributed applications** using FastAPI and Java while applying concepts from Operating Systems and Computer Networks. Strong foundation in Data Structures, Algorithms, and performance-oriented problem solving.

WORK EXPERIENCE

AI/ML Intern - Labmentix

Jun 2025 – Sep 2025

- Processed and analyzed **50k+ records**, improving model performance by **12%** through feature engineering and hyperparameter tuning.
- Built **end-to-end ML pipelines** covering data preprocessing, model training, validation, and performance benchmarking.
- Implemented evaluation metrics including **Precision, Recall, and F1-score** with automated experimentation workflows.

Associate - Next Tech Lab (SRM AP)

Apr 2025 – Jan 2026

- Contributed to projects involving **Machine Learning, NLP, and Generative AI** through research and technical discussions.
- Assisted in **model evaluation, optimization, and performance analysis** across multiple AI applications.
- Collaborated with cross-functional teams on prototyping, solution design, and technical reviews.

PROJECTS

TalentSync AI – Intelligent Resume Screening & Candidate Ranking Platform

[GitHub](#)

Python, FastAPI, MongoDB, NLP, FAISS, React, JWT, Docker, AWS EC2

- Built an **AI-powered recruitment platform** using **NLP, FAISS vector search**, and semantic matching for candidate screening.
- Implemented **384-dimensional embeddings, JWT authentication**, and **role-based access control** for recruiters and candidates.
- Developed **10+ FastAPI APIs**, integrated **MongoDB**, containerized with **Docker**, and deployed services on **AWS EC2**.

Real-Time Distributed Chat System

[GitHub](#)

Java, Socket Programming, Multithreading, MongoDB

- Built a **multithreaded distributed chat server** supporting **100+ concurrent clients** using TCP/IP sockets.
- Implemented **broadcast messaging, private channels**, and reliable real-time communication.
- Optimized concurrent processing, reducing message latency by **30%**, with **MongoDB-based persistent storage**.

Medical Report Analyzer (RAG System with Cloud Simulation)

[GitHub](#)

Python, FAISS, Transformers, FastAPI, CloudAnalyst

- Developed a **RAG-based medical report analyzer** using **FAISS**, embeddings, and NLP models for contextual explanations.
- Built scalable **FastAPI services** for semantic retrieval and AI-powered report processing.
- Simulated SaaS deployment in **CloudAnalyst**, optimizing load balancing for improved response time and cost efficiency.

Face Recognition Attendance System

[GitHub](#)

Python, OpenCV, DeepFace, Streamlit, SQLite

- Developed a real-time **face recognition attendance system** using **OpenCV** and **DeepFace** embeddings.
- Implemented **multi-face identification**, automated attendance logging, and persistent storage using **SQLite**.
- Built an interactive **Streamlit dashboard** with webcam integration for attendance tracking and management.

EDUCATION

2023 – Present **B.Tech in Computer Science & Engineering, SRM University, AP** **CGPA: 9.26 / 10.0**

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Machine Learning, Probability & Statistics, Computer Networks

Achievements: Ranked in **Top 10%** of cohort; consistently maintained high academic performance.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, SQL
- **AI/ML & Data Science:** TensorFlow, Scikit-learn, NLP, RAG, LLMs, Embeddings, Semantic Search, FAISS, LangChain
- **Backend Development:** FastAPI, Flask, REST APIs, JWT Authentication, Microservices
- **Databases:** MongoDB, MySQL, SQLite
- **Core Computer Science:** Data Structures & Algorithms, OOPs, Operating Systems, Computer Networks, DBMS
- **System Design & Distributed Systems:** Multithreading, Socket Programming, Distributed Systems
- **Cloud & Developer Tools:** AWS (EC2, S3), Docker, Git, GitHub, OpenCV, Power BI